Case Assignment Presentation

Creative Tooling

Lead Engineer

Part 1: Decoration Preview Service · Part 2: Tool Consolidation Strategy

30-Minute Presentation

Engineering · Design · Senior Leadership

Today's Agenda

Two-part presentation · 30 minutes total

Part 1

Decoration Preview Service

~10 minutes

- The problem & solution vision
- System architecture (C4)
- API design & security
- Scalability & design principles
- Implementation & summary

Part 2

Tool Consolidation Strategy

~20 minutes

- The challenge & vision
- Target architecture
- 5-phase 18-month roadmap
- Handling legacy tools & resistance
- AI strategy & success metrics

Part 1 · Decoration Preview Service

The Problem

Designers wait hours for 3D previews of their 2D artwork

Every preview request blocks the creative iteration cycle

Hours of waiting
Per preview request

Slow iteration
Feedback loops broken

Designer frustration
Creative momentum lost

Hours
Average wait time per preview

vs. seconds needed to stay in flow

Part 1 · Decoration Preview Service

Solution Vision

An async, cloud-native rendering service that never blocks the designer

Designers submit → system processes → result delivered — no waiting required

Async First

Submit & continue working. No blocking.

Cloud Native

AWS-native — scalable, resilient, managed.

Secure by Default

KMS encryption, pre-signed URLs, VPC isolation.

100× Scalable

100 to 10,000+ renders/ day with auto-scaling.

Part 1 · Architecture

C4 Context Diagram

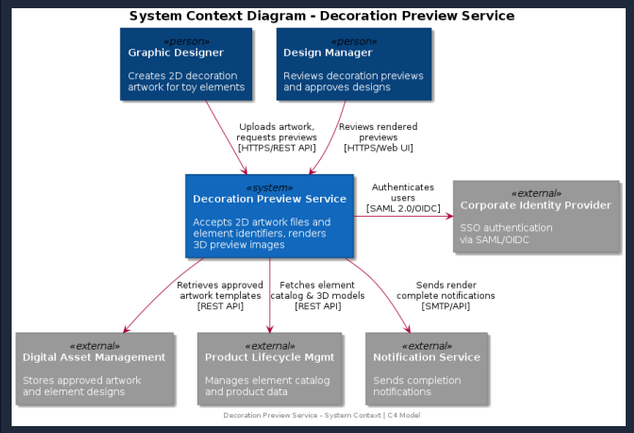
System Context — Who uses it and what does it connect to?

Primary Actors

- Graphic Designers — *Upload artwork, review previews*
- Design Managers — *Review & approve designs*

External Systems

- Digital Asset Mgmt — *Artwork templates*
- Product Lifecycle Mgmt — *3D element models*
- Corporate IdP — *SSO authentication*
- Notification Service — *Render completion*

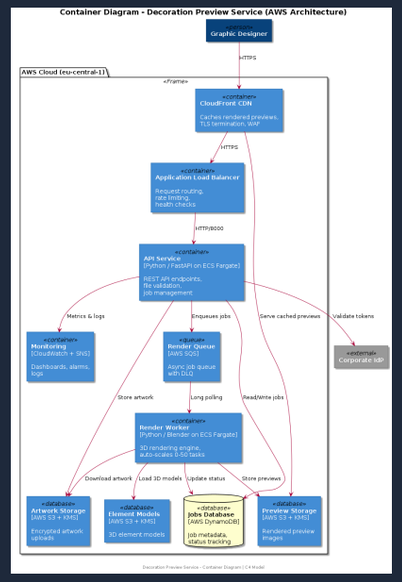


Part 1 · Architecture

C4 Container Diagram

AWS Infrastructure

- **CloudFront + WAF**
  Edge caching, DDoS protection, 1h TTL
- **API Service**
  FastAPI on ECS Fargate (2-10 tasks)
- **SQS Queue + DLQ**
  KMS encrypted, 10min visibility
- **Render Workers**
  0-50 ECS tasks, queue-driven scaling
- **S3 Buckets (×3)**
  Artwork, Elements, Renders
- **DynamoDB**
  Job tracking, GSI, TTL cleanup
- **CloudWatch**
  Dashboards, alarms, 30d logs



Part 1 · API Design

Five RESTful Endpoints

POST	`/api/v1/render` Submit artwork + element_id for rendering	202 Accepted
GET	`/api/v1/render/{job_id}/status` Poll render progress (0-100%)	200 OK
GET	`/api/v1/render/{job_id}/preview` Retrieve pre-signed S3 URL (1-hour expiry)	200 OK
GET	`/api/v1/elements` List available elements from catalog	200 OK
DELETE	`/api/v1/render/{job_id}` Cancel or delete a render job	204 No Content

Async Render Lifecycle

1. POST /render → validates file & element_id
2. Artwork stored in S3 (KMS encrypted)
3. Job created in DynamoDB (PENDING)
4. Message enqueued to SQS → 202 returned
5. **Designer keeps working — not blocked**
6. Worker polls SQS → renders → uploads to S3
7. GET /preview → pre-signed URL via CloudFront

Pydantic validates all requests before reaching the queue

Part 1 · Security

Defense in Depth — 6 Layers

Encryption (KMS)

All S3 buckets, DynamoDB, and SQS encrypted with AWS KMS Customer Managed Keys with annual rotation. TLS 1.2+ enforced on all connections. Pre-signed URLs with 1-hour expiry for preview access.

Authentication

API key auth via X-API-Key header (current). Roadmap: OIDC/SAML SSO with corporate IdP, JWT tokens with short expiry, per-element and per-team RBAC authorization.

Network Isolation

Render workers in private subnets with no public IP. VPC endpoints for S3, DynamoDB, SQS, and ECR — traffic never leaves AWS backbone. Security groups restrict API to port 8000 from ALB only.

Least-Privilege IAM

API role: S3 read/write artwork, DynamoDB read/write, SQS send. Render role: S3 read artwork, S3 write renders, SQS consume. CI/CD: ECR push and ECS deploy only.

WAF Protection

Rate limiting: 1,000 req/IP per 5 min. AWS Managed Rules for SQLi, XSS. Request body size limits. CloudFront absorbs DDoS at edge. Optional geo-blocking for corporate regions only.

Audit & Compliance

CloudTrail for all API calls. S3 access logging per artwork file. Structured JSON logs with correlation IDs, 30-day retention. VPC Flow Logs for network audit. SOC 2 ready.

STRIDE threat model applied — pre-release artwork classified as Confidential with High breach impact

Part 1 · Scalability

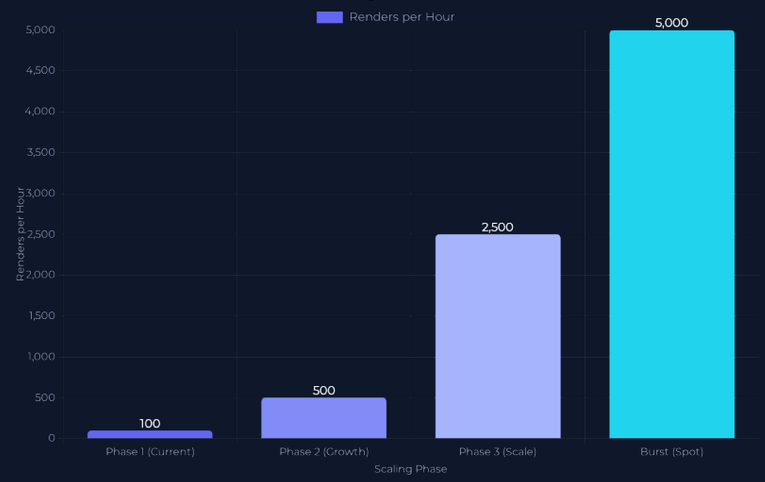
Scaling from 100 to 10,000+ Renders/Day

Performance Targets

API p99 latency	< 500ms
Render completion	< 120s
Concurrent renders	200+
Upload size limit	50 MB

Auto-Scaling Triggers

- API: CPU > 70% → scale 2-10 tasks
- Workers: Queue depth → scale 0-50 tasks
- Scale-to-zero when idle for 10 min
- Alarm: CPU >85%, Queue >100, DLQ >0



Phase 1	Phase 2	Phase 3
~$200/mo	~$800/mo	~$3-5K/mo
2 API + 1-5 workers	4 API + Spot workers + Redis	GPU + Multi-region + Batch

Part 1 · Engineering

10 Design Principles — Top 5

01

Async by Default

POST returns 202 immediately. SQS decouples submission from processing. A 5-min render never blocks a 200ms API call.

02

Separation of Concerns

API, JobStore, FileValidator, RenderWorker — each owns one domain. Infra split into 5 CDK stacks by domain.

03

Fail-Fast Validation

File type, size, MIME validated before queueing. Pydantic schema enforced at API layer. Invalid requests never reach SQS.

04

Idempotency

Unique job_id prevents duplicates. SQS visibility timeout stops double-processing. S3 keys are deterministic.

05

Design for Failure

DLQ captures 3× failed messages. ECS health checks restart unhealthy tasks. ALB routes around failures. Self-healing.

Applied Patterns: Factory · Repository · Strategy · Observer · CQRS · Event-Driven · IaC (CDK)

Part 1 · Implementation

What's Built & Why

56 Tests Passing | 88% Coverage | 5 CDK Stacks | 4 ADRs Documented

Component	Status	Notes
FastAPI Service	✅ Built	5 endpoints, Pydantic, async
SQS + DLQ	✅ Built	KMS encrypted, 3× retry
Render Workers	⚡ Stubbed	Pillow overlay (Blender in prod)
S3 / CloudFront	✅ Built	KMS, versioning, CDN delivery
DynamoDB + GSI	✅ Built	On-demand, TTL, PITR
GPU Rendering	⚡ Stubbed	Architecture ready, cost-controlled

Tech Stack Decisions

FastAPI
Async-native, auto-docs, Python ecosystem

SQS
Managed, durable, scales with ECS auto-scaling

ECS Fargate
Serverless containers, no EC2 management

CloudFront
Global CDN, integrates natively with S3

DynamoDB
Serverless, low-latency job state lookups

Part 1 · Summary

Key Decisions & Competencies

Technical Decisions

Async over sync
Frees designers from waiting — core UX win

AWS-native stack
Managed services reduce ops burden

Security first
Pre-release designs require defense-in-depth

Competencies Demonstrated

- System Architecture
- Security Design
- Scalability Planning
- API Design
- Async Patterns
- Cloud Native (AWS)
- Pragmatic Scoping
- Technical Communication

Part 2 · Tool Consolidation Strategy

A Fragmented Landscape

Current State

- Desktop App A — Tech Stack X
- Desktop App B — Tech Stack Y
- No shared auth
- Ad-hoc Script 1 — Python
- Ad-hoc Script 2 — Bash
- Knowledge silos
- Cloud Service A — Mature
- Cloud Service B — Early Stage
- No unified UI
- Legacy Tool — Undocumented
- Bus factor risk

Knowledge Silos
Expertise locked in individuals

No Shared Services
Auth, data, UI duplicated per tool

Bus Factor Risk
One departure = lost capabilities

Part 2 · Vision

Unified Platform-as-a-Service

One platform. Shared infrastructure, auth, data, and UI patterns.

Before

- Fragmented tools
- Duplicate infrastructure
- Knowledge silos
- No shared auth

→

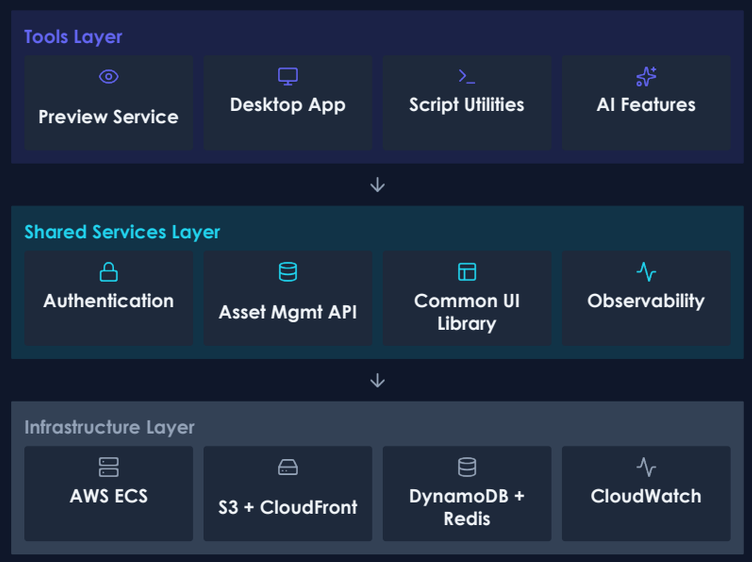
After

- Modular platform
- Shared services layer
- Collective ownership
- Unified authentication

Shared Services | **Common Auth** | **Consistent UI** | **Team Knowledge**

Part 2 · Architecture

Target Platform Architecture



- **Tools**
  Each tool is a module — plug in or out
- **Shared Services**
  Reused across all tools — no duplication
- **Infrastructure**
  Managed AWS services — low ops overhead

Part 2 · Strategy

Migration Philosophy

" No big bang. Designer-first. Always an escape hatch. "

Designer First

Every decision prioritises designer workflow continuity

Phased Approach

18-month roadmap — no single disruptive cutover

Always Rollback

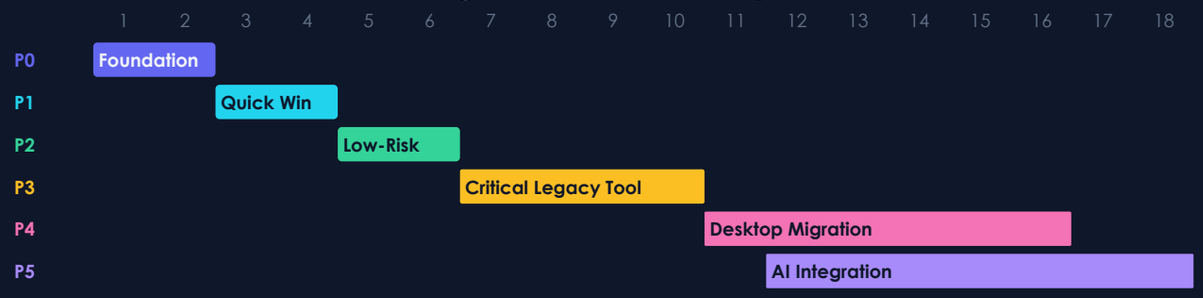
Every phase has an instant rollback mechanism

Zero Disruption

Blackout periods respected during product launches

Part 2 · Roadmap

18-Month Migration Roadmap



Phase 0	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Foundation	Quick Win	Low-Risk	Legacy Tool	Desktop	AI
Months 1–2	Months 3–4	Months 5–6	Months 7–10	Months 11–16	*Months 12–18*

Part 2 · Phases 0 & 1

Build Trust Before You Build Systems

Phase 0 Foundation

Months 1–2

Zero disruption. Build infrastructure and relationships.

- Establish designer advisory board
- Set up AWS infrastructure
- Create shared service layer
- Build trust with design team

Platform ready. Designers informed, not surprised.

Phase 1 Quick Win

Months 3–4

Launch the Preview Service. Prove platform value.

- Launch Preview Service as first platform tool
- Designers see immediate value
- Demonstrate ROI to stakeholders
- Build confidence in migration approach

First platform tool live. Credibility established.

Part 2 · Phases 2 & 3

Consolidation Begins

Phase 2 Low-Risk Migration

Months 5–6

Consolidate scripts and utilities. Low risk, visible progress.

- Migrate ad-hoc scripts to shared platform
- Standardize APIs and data formats
- Move to shared infrastructure

Multiple tools unified. Developer productivity up.

Phase 3 Critical Legacy Tool

Months 7–10

Highest Risk Phase

Reverse-engineer the undocumented tool. Zero regression.

- Capture input/output pairs from production
- Shadow mode — run old and new in parallel
- 3-stage rollout: 10% → 50% → 100%
- Instant rollback at every stage

Legacy tool replaced without a single regression.

Part 2 · Phase 3

Safe Legacy Tool Replacement

Poorly documented. Business-critical. Can't break it.

High Risk
No documentation, no tests, active usage

Prove it works before replacing it.

Capture
Record production input/output pairs as behavioral test cases

→

Shadow Mode
Run new implementation in parallel — compare outputs

→

Graduated Rollout
10% → 50% → 100% — monitor at each step

→

Validate & Retire
Only retire old tool after 100% validation

Part 2 · Phase 4

Desktop Migration (Months 11–16)

Phase 4 Desktop Migration

Migrate desktop apps without disrupting familiar workflows.

- Preserve familiar UI/UX patterns
- Gradual feature parity — no big bang
- Designer-led beta testing
- Rollback available throughout

Designers adopt voluntarily, not by mandate.

Change Management

Change Champions
Designers from the team lead adoption internally

Muscle Memory Preserved
Shortcuts, layouts, and workflows carried forward

Blackout Periods
No migrations during product launch windows

Part 2 · Phase 5

AI Strategy (Months 12–18)

AI as amplification, not replacement

Measured, tested, designer-vetoed

- A/B test every feature
- Measurable ROI per feature
- Designer veto on AI suggestions

Tier 1 — Low Risk, High Value

Low Risk

Smart search | Auto-tagging

Start here

Tier 2 — Medium Complexity

Medium Risk

Design variations | Anomaly detection

Proven demand

Tier 3 — Experimental

Measured

Text-to-design | Style transfer

Only if ROI proven

Part 2 · Change Management

Meeting Designers Where They Are

Designer Concern	Our Response
"I'll lose my workflow"	Preserve shortcuts, layouts, and muscle memory — we studied them first
"What if the new tool breaks?"	Instant rollback — old tool stays live throughout migration
"I wasn't consulted"	Designer advisory board from Day 1 — co-designed, not imposed
"Not during our launch!"	Blackout periods enforced — no migrations during key launch windows
"AI will replace me"	AI amplifies, never replaces. Designer veto on all AI suggestions

Part 2 · Success Metrics

How We Know It's Working

99.9%
Uptime SLA
Technical

NPS >50
Designer Satisfaction
Experience

30%
Faster Iteration Cycles
Business

40%
Cost Reduction
Business

Category	Metric	Target	Measurement
Technical	API response time	**< 2 seconds**	P99 latency (CloudWatch)
Technical	Render time	**< 30 minutes**	Job completion time
Experience	Weekly active users	**80% WAU**	Platform analytics
Experience	Voluntary adoption	**90% per phase**	Adoption tracking
Business	Iteration cycle speed	**+30% faster**	Design team survey
Business	Infrastructure cost	**–40%**	AWS Cost Explorer

Lead Engineer — Creative Tooling

Build with engineers.
Design with designers.
Deliver for the business.

- Technical Excellence
- Designer Empathy
- Strategic Delivery

Questions & Discussion

Thank you for your time